

Revisitation of Organoaluminum Reagents Affords a Versatile Protocol for C–X (X = N, O, F) Bond-Cleavage Cross-Coupling: A Systematic Study

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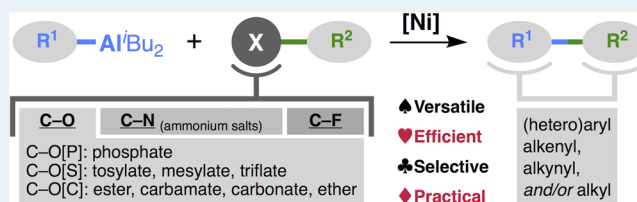
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Supporting Information

ABSTRACT: A revisit of organoaluminum reagents for cross-coupling reactions has opened up several types of C–C bond formation protocols through cleavage of phenolic/alcoholic C–O and C–F and ammonium C–N bonds. Catalyzed by the commercially available $\text{NiCl}_2(\text{PCy}_3)_2$ catalyst, these reactions proceed smoothly with a wide range of substrates and broad functional group compatibility, providing a versatile methodology for organoaluminum-mediated cross-coupling processes.

KEYWORDS: organoaluminum, cross-coupling, ether, organic fluoride compounds, ammonium salt, nickel catalyst, synthetic method



1. INTRODUCTION

Transition-metal (TM)-catalyzed cross-coupling reactions are some of the most important C–C bond-forming protocols and have been widely applied both in academic research and in industry.¹ In recent years, utilization of phenol/alcohol derivatives (C–O)^{2–5} and organic fluorides (C–F)^{6–9} as well as ammonium salts (C–N)^{10,11} as electrophiles in place of traditional halides (Cl, Br, and I) for TM-catalyzed cross-couplings has attracted much interest. These functional groups are widely present in nature or are important structural units for pharmaceuticals and functional materials. The use of the C–O/C–F/C–N electrophiles also has the advantage of avoiding halogen–metal exchange side reactions. Many elegant methodologies have already been developed, but issues of substrate diversity and selectivity are still challenging. Various types of C–O bond cleavage cross-coupling reactions have been reported, but in most cases, including the Kumada–Tamao-type,^{4a–k} Suzuki–Miyaura-type,^{4l–s,5d–i} Negishi-type,^{4t–v,5j} and others,^{5k–q} the reactions are often confined to naphthol-related substrates; in particular, many phenolic substrates exhibit low reactivity, even with excess amounts of organometallic nucleophiles and/or under harsh conditions. For the pioneering work on C–F bond cleavage cross-coupling, especially with organoboron⁸ and organozinc,⁹ reactions are limited to aryl fluorides with extremely electron deficient nature or with *ortho* group assistance, and electron-rich aryl compounds are invariably less reactive. Therefore, efficient and versatile methods with broad substrate scope and high selectivity are still needed.

Here we report an efficient protocol for organoaluminum-mediated cross-coupling reactions with a variety of phenol/

alcohol derivatives, organic fluorides, and ammonium salts, providing a new methodology for C–C bond formation via C–O/F/N bond cleavage. The initial use of organoaluminum¹² in TM-catalyzed cross-coupling reactions can be traced back to Negishi's reports as early as in 1976,¹³ but in subsequent decades, organoaluminum has been largely neglected in favor of other organometallics (Mg, Zn, B, Sn, etc.).¹ A few cross-coupling reactions with organoaluminums have been reported, though most of them focused on reactions with organic halides.¹⁴ Very recently, Ni-catalyzed cross-coupling between aryl methyl ether (ArOMe) and trialkylaluminum (R_3Al) was reported by Chatani, Tobisu, et al.,^{4u} and Rueping et al.^{4t} However, organoaluminum reagents offer many synthetic advantages, including low cost, ready availability, low toxicity, and in particular, exceptional Lewis acidity. We have also found that organoaluminum reagents exhibit low halogen–metal exchange ability.^{14b,15} This is important, because halogen–metal exchange usually occurs as a side reaction in cross-coupling, especially in TM-catalyst-free single-electron-transfer (SET)-induced direct coupling between organic halides and organometallic reagents, such as Grignard reagents and organozinc compounds.¹⁶ In 2015, we serendipitously discovered a direct cross-coupling reaction between organic halides and organoaluminums without any external catalyst.^{14b} This reaction proceeds smoothly without formation of detectable amounts of dehalogenated side product and is applicable to a wide range of organic halides, including aryl/

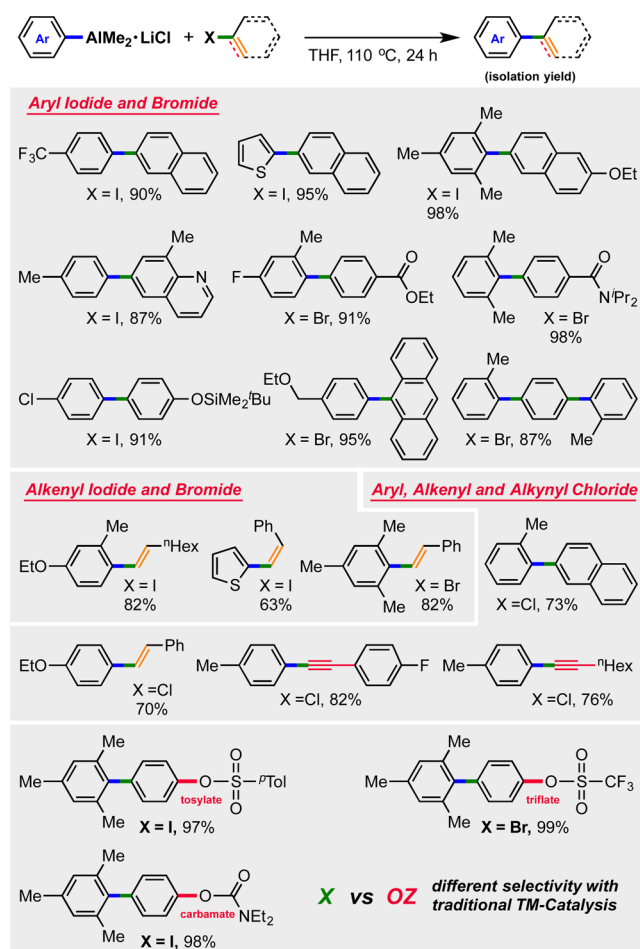
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alkenyl/alkynyl halides, with broad functional group compatibility (Scheme 1). Further, when aromatic iodides or bromides

Scheme 1. Direct Cross-Coupling between Aluminum Reagents and Various Types of Organic Halides^a



^aResults from ref 14b

bearing tosylate, triflate, and carbamate groups were employed, the reactions took place specifically at the halogen sites in excellent yield, even for very bulky aluminum reagents with bromide and triflate.¹⁷

Encouraged by the advantages of high reactivity and selectivity, we made a systematic study of organoaluminum-mediated cross-coupling reactions via inert bond (C–O, C–N, C–F) cleavage. Notably, all these reaction proceed under mild conditions with the simple and commercially available $\text{NiCl}_2(\text{PCy}_3)_2$ catalyst, and substrates having non-naphthalene structures, including electron-rich and/or bulky aromatic rings that show little or no reactivity in previously reported protocols, reacted smoothly to give the desired products in high yields. Therefore, this methodology provides an efficient, selective, and versatile synthetic protocol.

2. RESULTS AND DISCUSSION

2.1. Ni-Catalyzed Cross-Coupling between Organoaluminum Reagents and Phenol/Alcohol Derivatives. In recent years, utilization of carboxylates (ester, carbamate, etc.) and ethers as substrates for cross-couplings via C–O bond cleavage has attracted much attention, owing to their easy

accessibility, low cost, and high bench stability.^{2–5} Using **2a**^{carbamate} as a model substrate, initial studies aimed at developing coupling conditions revealed that the combination of PhAl^iBu_2 (**1a**) (prepared from PhMgCl ^{18,19} and ClAl^iBu_2 ^{14d}) as the aluminum reagent, toluene as the solvent, and $\text{NiCl}_2(\text{PCy}_3)_2$ as the catalyst²⁰ was a suitable starting point for examination of the substrate scope of the present cross-coupling reaction (Table 1). Under optimized conditions, we

Table 1. Modifications of Reaction Conditions^a

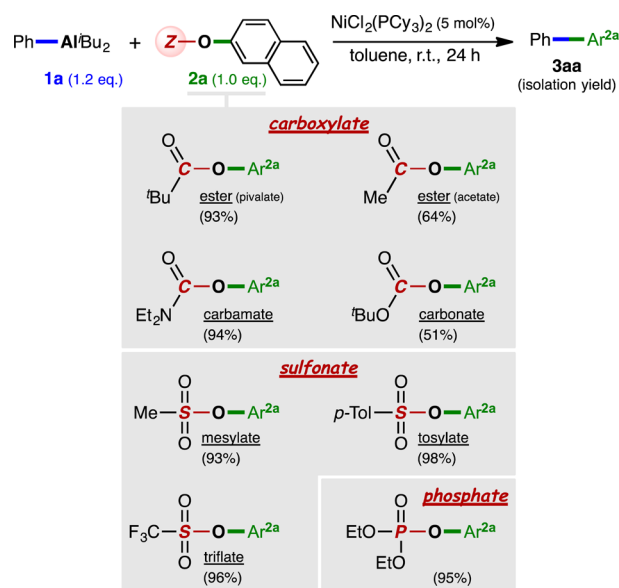
Entry	Ph-[Al] (1a)	(PhLi + AlMe ₃)	2.0 eq.	Solvent	Temperature	GC yield ^b
1	Ph-AlMe ₂ Li	(PhLi + AlMe ₃)	2.0 eq.	toluene	90 °C	trace
2	Ph-AlMe ₂	(PhLi + AlMe ₂ Cl)	2.0 eq.	toluene	90 °C	18%
3	Ph-AlMe ₂	(PhLi + AlMe ₂ Cl)	2.0 eq.	THF	90 °C	57%
4	Ph-Al ⁱ Bu ₂	(PhLi + Al ⁱ Bu ₂ Cl)	2.0 eq.	THF	70 °C	90%
5	Ph-Al ⁱ Bu ₂	(PhLi + Al ⁱ Bu ₂ Cl)	2.0 eq.	THF	50 °C	64%
6	Ph-Al ⁱ Bu ₂	(PhMgCl + Al ⁱ Bu ₂ Cl)	2.0 eq.	THF	50 °C	81%
7	Ph-Al ⁱ Bu ₂	(PhMgCl + Al ⁱ Bu ₂ Cl)	2.0 eq.	toluene	50 °C	>95%
8	Ph-Al ⁱ Bu ₂	(PhMgCl + Al ⁱ Bu ₂ Cl)	1.5 eq.	toluene	r.t.	>95%
9	Ph-Al ⁱ Bu ₂	(PhMgBr + Al ⁱ Bu ₂ Cl)	1.5 eq.	toluene	r.t.	40%

^aConditions: **1a** (0.75 mmol, 1.5 equiv or 1.0 mmol, 2.0 equiv) was treated with **2a** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) in toluene (3 mL) at room temperature to 90 °C under argon. ^bInternal standard for GC yield: *n*-dodecane.

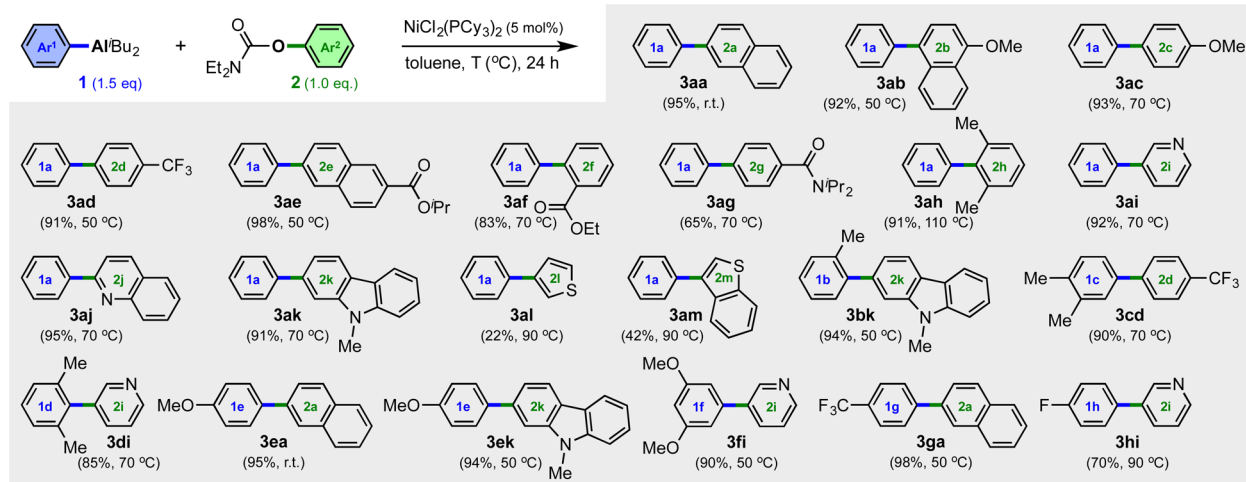
evaluated the scope of the coupling reaction using several types of C–O substrates **2a**, including esters, carbonate, tosylate, mesylate, triflate, and phosphate. In most cases, the reaction proceeded smoothly even with 1.2 equiv of **1a**, and the coupling product **3aa** was obtained in high yield (Scheme 2).

Taking the carbamate as a representative C–O moiety, we next investigated the scope and limitations of the present cross-coupling of **2** with various arylaluminum reagents **1** (Scheme 3). The results can be summarized as follows. (1) Aryl

Scheme 2. Cross-Coupling between Aluminum Reagent **1a and Various Naphthol Derivatives **2a** in the Presence of Ni Catalyst^a**



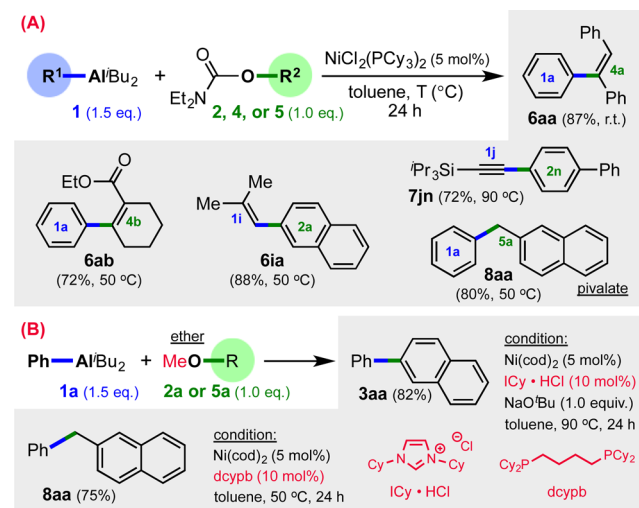
^aConditions: **1a** (0.6 mmol, 1.2 equiv) was treated with **2a** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) in toluene (3 mL) at room temperature under argon.

Scheme 3. Cross-Coupling between Aluminum Reagent 1 and Phenol Carbamates 2 in the Presence of Ni Catalyst^{a,b}

^aConditions: **1** (0.75 mmol, 1.5 equiv) was treated with **2** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) in toluene (3 mL) at room temperature to 90 °C under argon. ^bIsolated yields.

carbamates: the reactivity of many phenol carbamates appeared to be lower than that of 2-naphthol derivatives **2a** at room temperature, but heating at higher temperature improved the reactivity, affording the biaryl products in high yields in most cases. (2) Influence of electronic properties of the aryl carbamates **2**: a variety of functional groups, such as methoxy (**2b,c**), trifluoromethyl (**2d**), ester (**2e**), amide (**2g**), and heterocycles (**2i–k**), could be employed, and their electronic properties had little influence on the reactivity, except for thiophene-containing substrates (**2l,m**; unreacted substrates remained). (3) Aromatic aluminum reagents: while electron-rich arylaluminums showed higher reactivity in comparison to electron-deficient species (**1b,c,e,f** vs **1g,h**), gentle heating in the reactions of **1g,h** bearing electron-withdrawing groups also led to satisfactory yields. (4) Steric factors: steric bulkiness of both aromatic aluminum (**1b** and even **1d**) and aryl carbamate (**1f**) did not greatly affect the reactivity, though high temperature was needed to achieve a good yield in the reaction with very bulky aryl carbamate (**2h**). In general, existing coupling protocols via C–O bond cleavage, such as Kumada–Tamao,^{4a–k} Suzuki–Miyaura,^{4l–s,Sd–i} and Negishi-types,^{4t–v,Sj} are often confined to naphthol-related substrates; in particular, many phenolic substrates exhibit low reactivity, even under harsh conditions. In sharp contrast, the present method is broadly applicable to a variety of phenol derivatives, even those bearing electron-donating or bulky groups, opening up new potential applications of organoaluminum reagents in this area.

In addition to aromatic C–O substrates, vinylic (enolic) **4** and even benzyl alcohol ($\text{C}(\text{sp}^3)\text{–O}$)^{2c} derivatives **5** were also available for this reaction. Representative results of the coupling of these substrates are summarized in Scheme 4A. Both acyclic (**4a**) and cyclic substrates (**4b**) smoothly underwent C–(alkenyl)–O bond cleavage to give the desired coupling products **6aa,ab** in good yields. Alkenyl (**1i**)- and alkynylaluminum (**1j**) reagents could also be utilized for this cross-coupling, affording the corresponding products **6ia** and **7jn** without difficulty. Therefore, this reaction provides a simple and direct method for the regiocontrolled synthesis of multiply substituted olefins/alkynes. Furthermore, pivalate **5a**^{pivalate} reacted with **1a** to afford diarylmethane **8aa** in 80% yield. On the other hand, cleavage/functionalization of the ethereal C–O

Scheme 4. (A) Cross-Coupling between Aluminum Reagent 1 and Carbamate 2, 4, or 5 To Afford C_{vinylic}–C_{phenyl}/C_{alkynyl}–C_{phenyl}/C_{phenyl}–C_{alkyl} Bonds and (B) Cross-Coupling between Aluminum Reagent 1a and Ether 2a or 5a^{a,b}

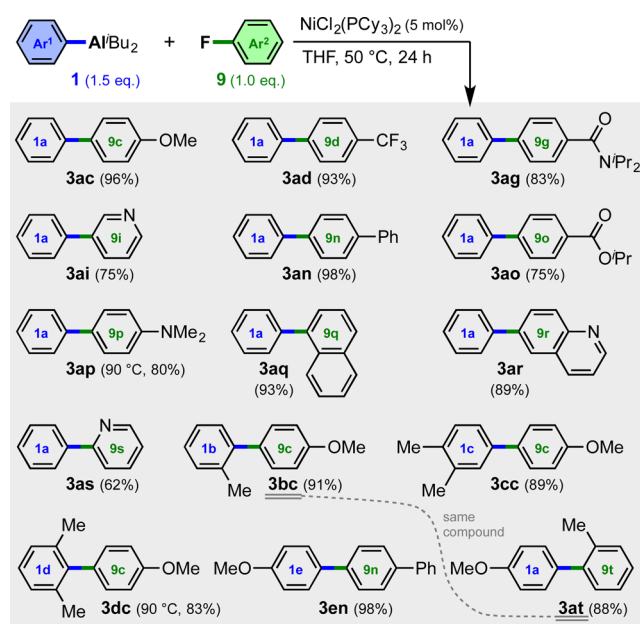
^aConditions: **1** (0.75 mmol, 1.5 equiv) was treated with **2**, **4**, or **5** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) or $\text{Ni}(\text{cod})_2$ (5 mol %)/ligand (10 mol %) in toluene (3 mL) at room temperature to 90 °C under argon. ^bIsolated yields.

bond is much more difficult due to its inertness.^{2–4} In fact, simple application of the current protocol to 2-methoxynaphthalene **2a**^{OMe} proved to be less effective. For example, heating excess amounts of **1a** (2.0 equiv) and **2a**^{OMe} with $\text{NiCl}_2(\text{PCy}_3)_2$ at high temperature (110 °C) gave **3aa** in only 59% yield, but the yield increased to 82% when the reaction mixture was heated in THF in the presence of $\text{Ni}(\text{cod})_2$, imidazolium ligand (ICy), and NaO^tBu as an additive (Scheme 4B).^{4r} Interestingly, benzyl ether **6a**^{OMe}, which has higher reactivity than aryl ether,^{2c} reacted smoothly at 50 °C in the presence of $\text{Ni}(\text{cod})_2$ and dcybp to give **8aa** in 75% yield (Scheme 4B).

2.2. Ni-Catalyzed Cross-Coupling between Organoaluminum Reagents and Organic Fluorides. With the success of cross-coupling through C–O bond cleavage, we

turned our attention to C–F bond cleavage⁶ as the next target for utilization of aluminum reagents. The C–F bond shows high stability to a wide range of reagents and metals. Several methods for C–C bond formation via cross-coupling of organic fluorides have been reported to date, mostly involving Grignard reagent,⁷ organoboron,⁸ and organozincs.⁹ Many pioneering studies on cross-coupling with organic fluorides were limited to aryl fluorides with extremely electron-deficient nature or with *ortho* group assistance, especially for Suzuki–Miyaura-type and Negishi-type reactions. A more general protocol for Suzuki–Miyaura coupling with a wide range of aryl fluorides was recently developed by Tobisu and Chatani, based on efficient promotion by additional Lewis acids.⁸¹ In this section, we demonstrate the remarkable reactivity of aluminum reagents **1** toward a variety of aryl fluorides **9**, ranging from electron-rich to electron-deficient aromatic rings, in the absence of an *ortho* directing group (Scheme 5). It is known that electron-rich aryl

Scheme 5. Cross-Coupling between Arylaluminum Reagent **1** and Aryl Fluoride **9** Catalyzed by Ni Catalyst^{a,b}

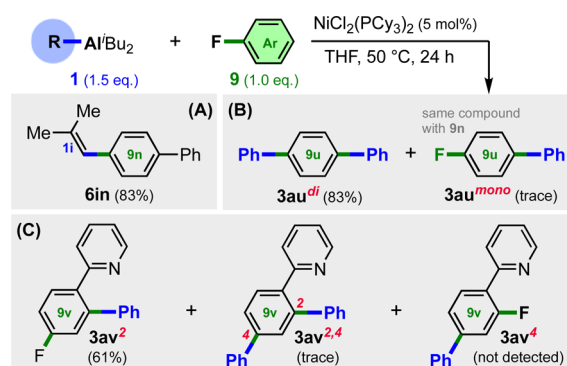


^aConditions: **1** (0.75 mmol, 1.5 equiv) was treated with **9** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) in THF (3 mL) at 50–90 °C under argon. ^bIsolated yields.

fluorides are normally less reactive electrophiles for the Suzuki–Miyaura-type,⁸ Negishi-type,⁹ and even some cases of Kumada–Tamao-type⁷ couplings, whereas by using the current protocol, the reaction with aryl fluoride **9c** bearing an electron-donating OMe group took place smoothly, even with the sterically demanding arylaluminum reagent **1d** (albeit at higher temperature). Similarly, the reaction with electron-rich **9p** also gave a good yield at higher temperature. Further, various polar functional groups are well tolerated, such as CF_3 (**9d**),²¹ amide (**9g**), and ester (**9o**), as well as heterocycles (**9i,r,s**), some of which are sensitive to metal reagents and base. The reaction also proceeded well at the sterically hindered *alpha*-position of the naphthalene ring (**9q**) and the *ortho*-substituted phenyl ring (**9t**).

Although alkenylaluminum reagents could also be utilized in this reaction (Scheme 6A), alkyl and alkynyl aluminums gave only small amounts of the coupling product. When 1,4-

Scheme 6. Ni-Catalyzed Cross-Coupling of (A) Alkenylaluminum **1i** with Aryl Fluoride **9n** and of Phenylaluminum **1a** (B) with 1,4-Difluorobenzene **9u** and (C) with 2,4-Difluorobenzene **9v** Bearing a Directing Group^{a,b}

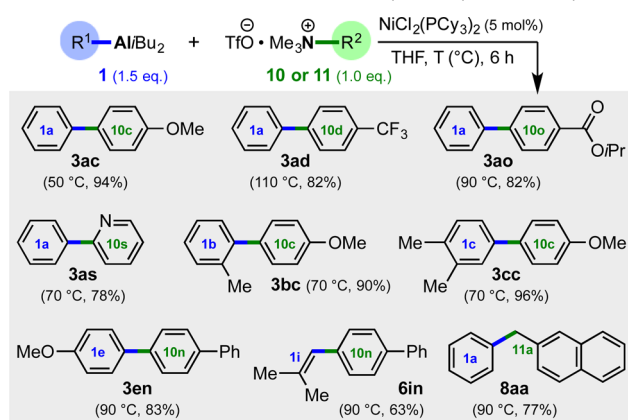


^aConditions: **1** (0.75 mmol, 1.5 equiv) was treated with **9** (0.5 mmol, 1 equiv) in the presence of $\text{NiCl}_2(\text{PCy}_3)_2$ (5 mol %) in THF (3 mL) at 50 °C under argon. ^bIsolated yields.

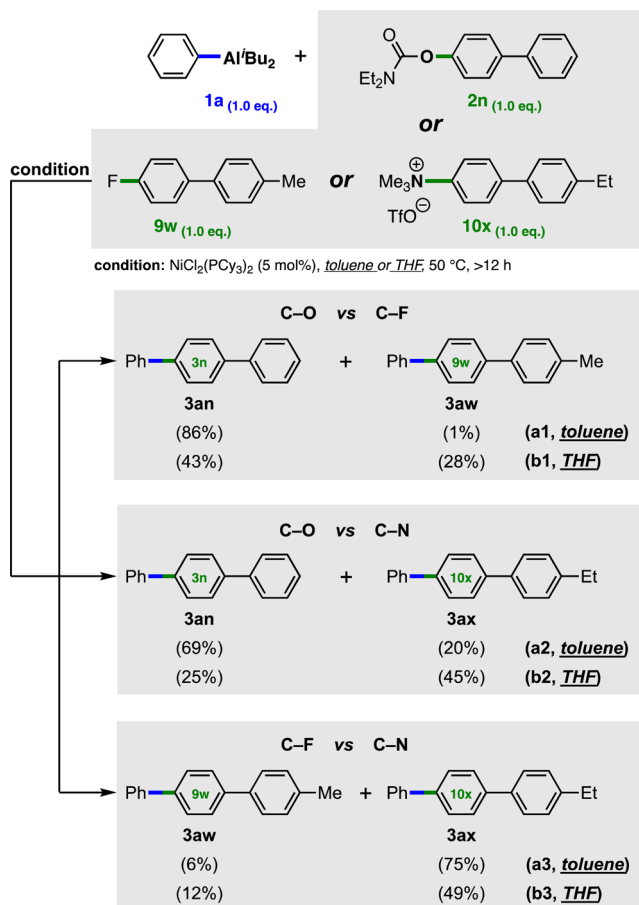
difluorobenzene **9u** was employed (Scheme 6B), both C–F bonds were involved in the reaction and the double cross-coupling product was obtained in 83% yield. On the other hand, **9v** underwent “single” cross-coupling regioselectively, and only one C–F bond adjacent to the directing group was arylated (Scheme 6C). This result suggests the feasibility of a selective functionalization strategy for multifluoroarenes.

2.3. Ni-Catalyzed Cross-Coupling between Organoaluminum Reagents and Arylammonium Salts. Ammonium salts can easily be prepared directly from the corresponding amines and show much higher C–N reactivity.¹⁰ Hence, TM-catalyzed cross-coupling reactions with ammonium salts as the electrophiles have attracted much attention in recent years.¹¹ After extensive experimentation, we found that heating in THF in the presence of Ni catalyst provided satisfactory reactivity with organoaluminum reagents. Arylammonium salts **10** substituted with electron-donating and electron-withdrawing groups reacted with arylaluminum reagent **1** to afford the corresponding coupling products in good yields (Scheme 7). Alkylammonium salt **11a** as well as alkenylaluminum reagent **1i** were also available for this reaction.

2.4. Comparison of Reactivities of C–O/C–F/C–N Substrates in Ni-Catalyzed Cross-Coupling with Organoaluminum Reagents. Finally, to investigate the chemoselectivity of this coupling, the reactivities of the cleavage of C–O (carbamate), C–F, and C–N (ammonium) bonds were examined (Scheme 8). Interestingly, we found that the chemoselectivity of the present coupling system was dependent upon the reaction solvent. In toluene, both the C–O bond in carbamate and the C–N bond in ammonium salt showed much higher reactivity than the C–F bond (Scheme 8, a1–a3). In comparison with the case of carbamate C–O bond, the decrease in reactivity of ammonium salt in toluene is probably due to low solubility (Scheme 8, a2). On the other hand, in THF, the C–N bond in ammonium salt showed slightly higher reactivity than the carbamate C–O bond and is preferentially cleaved over the C–F bond (Scheme 8, b1–b3). Thus, C–F bond cleavage was the most sluggish process both in toluene and in THF. Although the selectivity is poor in THF, it is noteworthy that the competitive reactions in toluene for C–O vs C–F (Scheme 8, a1) and C–F vs C–N (a3) were highly

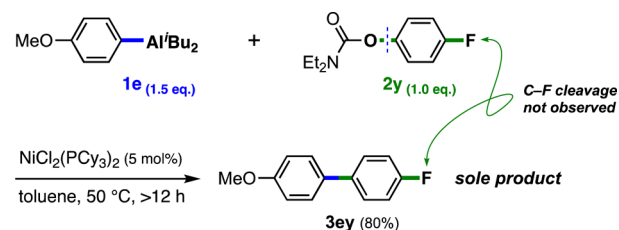
Scheme 7. Cross-Coupling between Aluminum Reagent 1 and Ammonium Salt 10 or 11 Catalyzed by Ni Catalyst^{a,b}

^aConditions: **1** (0.75 mmol, 1.5 equiv) was treated with **10** or **11** (0.5 mmol, 1 equiv) in the presence of $NiCl_2(PCy_3)_2$ (5 mol %) in THF (3 mL) at 50–110 °C under argon. ^bIsolated yields.

Scheme 8. Reactivity and Chemoselectivity in Ni-Catalyzed Cross-Coupling with Aryl Aluminum Reagents **1** via C–O/C–N/C–F Bond Cleavages^{a,b}

^aConditions: **1a** (0.5 mmol, 1.0 equiv) was treated with a 1/1 mixture (0.5 mmol, 1 equiv for both components) of **3n/9w**, **3n/10x**, and **9w/10x** in the presence of $NiCl_2(PCy_3)_2$ (5 mol %) in toluene (3 mL) or THF (3 mL) at 50 °C under argon. ^bIsolated yields.

selective. We also performed a representative intramolecular competition reaction of **2y** with **1e** in toluene (Scheme 9). As

Scheme 9. Chemoselective Coupling between **1e** and **2y**^{a,b}

^aConditions: **1e** (0.75 mmol, 1.5 equiv) was treated with **2y** (0.5 mmol, 1 equiv) in the presence of $NiCl_2(PCy_3)_2$ (5 mol %) in toluene (3 mL) at 50 °C under argon. ^bIsolated yields.

in the intermolecular reaction, although carbamate **2y** contains both C–F and the C–O moieties, **3ey** was obtained in good yield as the sole product via C–O cleavage. The characteristic selectivity of the present coupling reaction in solvents with different coordinating ability and polarity provides some mechanistic insights. Further studies to elucidate the reaction pathways with the help of theoretical and spectroscopic studies are in progress in our laboratory.

3. CONCLUSION

In summary, we have established new protocols for cross-coupling reactions utilizing organoaluminum reagents and transition-metal catalysts. The present re-examination has greatly expanded the range of applicability of organoaluminums in cross-coupling reactions. The use of commercially available Ni catalyst and easily accessible organoaluminum resources as a coupling partner is highly attractive. (1) Taking advantage of the Rieke method, various organoaluminum reagents could be readily prepared from $RMgCl$ and directly used for coupling reactions without the need for isolation/purification, in contrast to the case of organoboron reagents. (2) Aluminum reagents show broad compatibility with various active functional groups, especially in C–O substrates, that might be poorly compatible with sole use of Grignard reagent. (3) The present methods demonstrate the superior reactivity of aluminum reagents, especially toward non-naphthol-related C–O substrates that bear electron-donating or bulky groups, as well as electron-rich aryl fluorides (C–F), which show little or no reactivity in existing coupling protocols with Zn, B, or Mg reagents. In general, the current protocols of aluminum reagents show favorable reaction characteristics, mild conditions, wide substrate scope, and broad functional group compatibility, suggesting that the new reactions should be widely useful in organic synthesis. Detailed mechanistic studies and work to extend the present methodology to other important carbon–carbon/carbon–heteroatom bond-forming reactions are in progress.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b01058.

Experimental procedures, characterization data, and ¹H and ¹³C NMR spectra for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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